

## 1 Summary

Model (Run-name): model1 (test)

Iteration: 28

**TAC: 154882.24055074638**

Took: 7.157219886779785s

Total Time: 56.608795404434204s

Epsilons:

- $\varepsilon_{LMTD}$ : 1e-06
- $\varepsilon_{Balancing}$ : 1e-06
- $\varepsilon_{Area}$ : 1e-06
- $\varepsilon_{Beta}$ : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

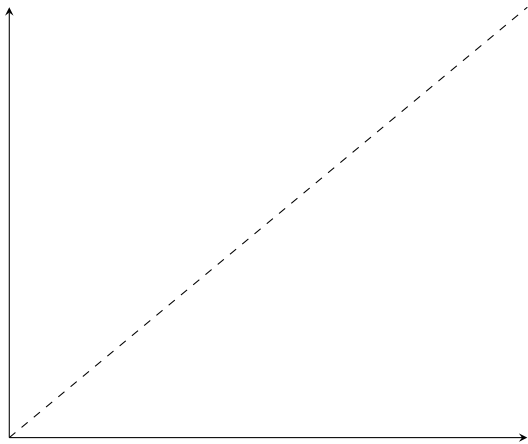
Gamsworld Solution: [http://www.gamsworld.org/minlp/minlplib2/html/heatexch\\_gen1.html](http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html)

## 2 Results

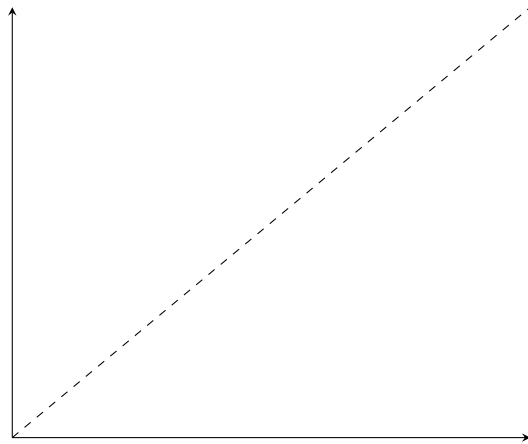
| Variable          | Value              | Variable           | Value                | Variable            | Value              |
|-------------------|--------------------|--------------------|----------------------|---------------------|--------------------|
| $z_{1,1,1}$       | 1.0                | $RecLMTD_{1,1,1}$  | 0.05184205328699354  | $b_{1,1,1}^{H,out}$ | 2395.852553577473  |
| $z_{1,2,2}$       | 1.0                | $RecLMTD_{1,2,2}$  | 0.01809391399811904  | $b_{1,2,2}^{H,out}$ | 3857.856293919187  |
| $z_{2,1,2}$       | 1.0                | $RecLMTD_{2,1,2}$  | 0.028081985276110513 | $b_{2,1,2}^{H,out}$ | 4056.182923296087  |
| $z_{cu,1}$        | 1.0                | $RecLMTD_{cu,1}$   | 0.014733536234918122 | $b_{1,1,1}^{C,in}$  | 8561.78444087873   |
| $z_{cu,2}$        | 1.0                | $RecLMTD_{cu,2}$   | 0.009546972511543995 | $b_{1,2,2}^{C,in}$  | 4550.0             |
| $z_{hu,1}$        | 1.0                | $RecLMTD_{hu,1}$   | 0.022468643315065755 | $b_{2,1,2}^{C,in}$  | 4171.264629527628  |
| $A_{1,1,1}^\beta$ | 71.62767096549581  | $q_{1,1,1}$        | 692.1437060808369    | $b_{1,1,1}^{C,out}$ | 9253.928146959566  |
| $A_{1,2,2}^\beta$ | 70.56626459266522  | $q_{1,2,2}$        | 1950.0               | $b_{1,2,2}^{C,out}$ | 6500.0             |
| $A_{2,1,2}^\beta$ | 135.25996987589937 | $q_{2,1,2}$        | 2411.784440878792    | $b_{2,1,2}^{C,out}$ | 6583.0490704064205 |
| $A_{cu,1}^\beta$  | 4.600843803756838  | $q_{cu,1}$         | 157.85629391918576   |                     |                    |
| $A_{cu,2}^\beta$  | 37.95085342166671  | $q_{cu,2}$         | 1988.2155591212213   |                     |                    |
| $A_{hu,1}^\beta$  | 13.363827419897227 | $q_{hu,1}$         | 496.07185304041366   |                     |                    |
| $A_{1,1,1}$       | 71.62767096549581  | $t_{1,1}^{(H)}$    | 650.0                |                     |                    |
| $A_{1,2,2}$       | 70.56626459266522  | $t_{1,2}^{(H)}$    | 580.7856293919187    |                     |                    |
| $A_{2,1,2}$       | 135.25996987589937 | $t_{1,3}^{(H)}$    | 385.7856293919186    |                     |                    |
| $A_{cu,1}$        | 4.600843803756838  | $t_{2,1}^{(H)}$    | 590.0                |                     |                    |
| $A_{cu,2}$        | 37.95085342166671  | $t_{2,2}^{(H)}$    | 590.0                |                     |                    |
| $A_{hu,1}$        | 13.363827419897227 | $t_{2,3}^{(H)}$    | 469.41077795606105   |                     |                    |
| $dt_{1,1,1}$      | 33.0714568693611   | $t_{1,1}^{(C)}$    | 616.9285431306391    |                     |                    |
| $dt_{1,1,2}$      | 10.0               | $t_{1,2}^{(C)}$    | 570.7856293919192    |                     |                    |
| $dt_{1,2,2}$      | 80.78562939191872  | $t_{1,3}^{(C)}$    | 410.0                |                     |                    |
| $dt_{1,2,3}$      | 35.78562939191872  | $t_{2,1}^{(C)}$    | 500.0                |                     |                    |
| $dt_{2,1,2}$      | 19.21437060808333  | $t_{2,2}^{(C)}$    | 500.0                |                     |                    |
| $dt_{2,1,3}$      | 59.4107779560612   | $t_{2,3}^{(C)}$    | 350.0                |                     |                    |
| $dt_{cu,1}$       | 65.78562939191858  | $t_{1,1,1}^H$      | 504.30895275685174   |                     |                    |
| $dt_{cu,2}$       | 149.41077795606105 | $t_{1,2,2}^H$      | 385.7856293919187    |                     |                    |
| $dt_{hu,1}$       | 63.07145686936091  | $t_{2,1,2}^H$      | 370.0                |                     |                    |
| $f_{1,1,1}^H$     | 4.7507634763974    | $t_{1,1,1}^C$      | 616.9285431305755    |                     |                    |
| $f_{1,2,2}^H$     | 10.0               | $t_{1,2,2}^C$      | 500.0                |                     |                    |
| $f_{2,1,2}^H$     | 10.962656549448859 | $t_{2,1,2}^C$      | 646.6953103605483    |                     |                    |
| $f_{1,1,1}^C$     | 15.0               | $b_{1,1,1}^{H,in}$ | 3087.99625965831     |                     |                    |
| $f_{1,2,2}^C$     | 13.0               | $b_{1,2,2}^{H,in}$ | 5807.856293919187    |                     |                    |
| $f_{2,1,2}^C$     | 10.173816169579581 | $b_{2,1,2}^{H,in}$ | 6467.967364174907    |                     |                    |

### 3 *RecLMTD* Tangent Points

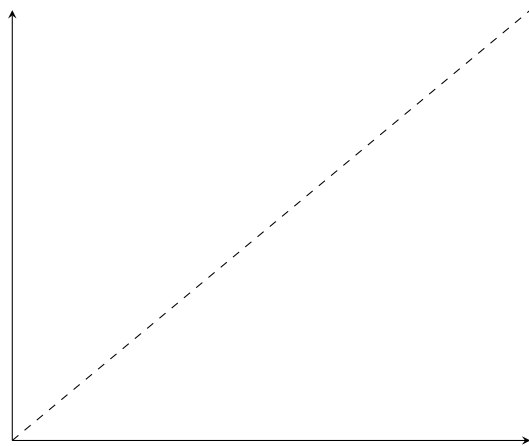
- $(1, 1, 1)$ 
  - Value: 0.051842
  - Error: 0.000000597738915
  - Relative error: 0.000011529867846
- $(1, 2, 2)$ 
  - Value: 0.018094
  - Error: 0.000000590367740
  - Relative error: 0.000032626908584
- $(2, 1, 2)$ 
  - Value: 0.028082
  - Error: 0.000000554015705
  - Relative error: 0.000019728120013
- $cu, 1$ 
  - Value: 0.014734
  - Error: 0.000000295134945
  - Relative error: 0.000020031106470
- $cu, 2$ 
  - Value: 0.009547
  - Error: 0.000000902338311
  - Relative error: 0.000094506717505
- $hu, 1$ 
  - Value: 0.022469
  - Error: 0.000000005130830
  - Relative error: 0.000000228355093



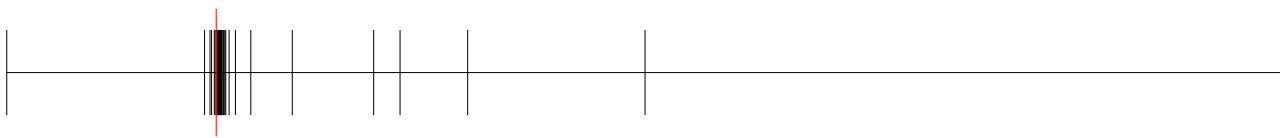
(a) *RecLMTD* - (1, 1, 1)



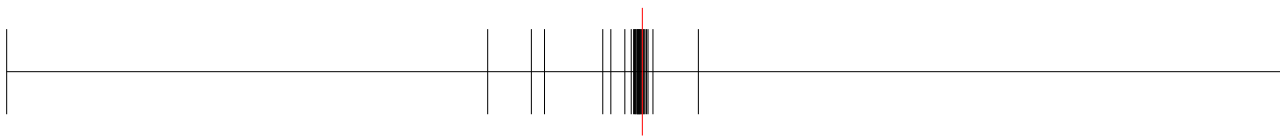
(b) *RecLMTD* - (1, 2, 2)



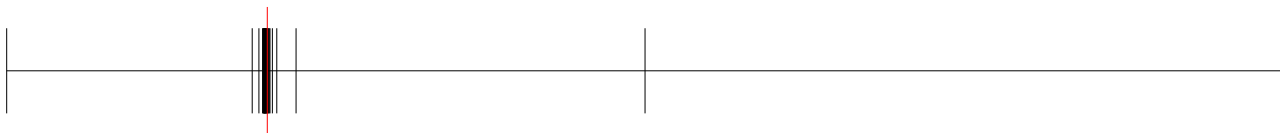
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

## 4 Balancing Breakpoints

### 4.1 $b^{H,\text{in}}$

- (1, 1, 1)
  - Value: 3087.99625965831
  - Absolute error: 0.0
  - Relative error: 0.0
- (1, 2, 2)
  - Value: 5807.856293919187
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 1, 2)
  - Value: 6467.967364174907
  - Absolute error: 8.003553375601768e-11
  - Relative error: 1.2374140011794647e-14

### 4.2 $b^{H,\text{out}}$

- (1, 1, 1)
  - Value: 2395.852553577473
  - Absolute error: 0.0
  - Relative error: 0.0
- (1, 2, 2)
  - Value: 3857.856293919187
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 1, 2)
  - Value: 4056.182923296087
  - Absolute error: 9.549694368615746e-12
  - Relative error: 2.354354956175302e-15

### 4.3 $b^{C,\text{in}}$

- (1, 1, 1)
  - Value: 8561.78444087873
  - Absolute error: 5.820766091346741e-11
  - Relative error: -6.798543144294921e-15
- (1, 2, 2)
  - Value: 4550.0
  - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
  - Value: 4171.264629527628
  - Absolute error: 0.0
  - Relative error: 0.0

#### 4.4 $b^{C,out}$

- (1, 1, 1)
  - Value: 9253.928146959566
  - Absolute error: 9.349605534225702e-10
  - Relative error: 1.0103391106725328e-13
- (1, 2, 2)
  - Value: 6500.0
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 1, 2)
  - Value: 6583.0490704064205
  - Absolute error: 3.6898650689881833
  - Relative error: 0.0005608243833221359



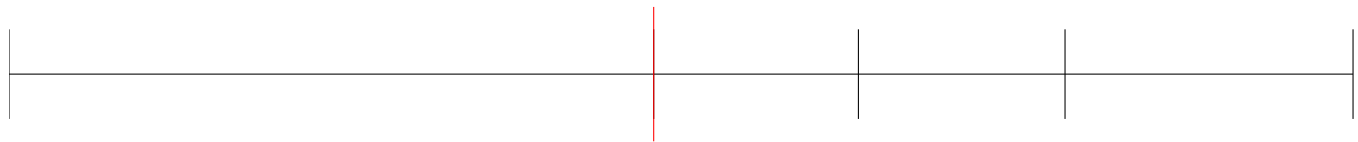
(a)  $t^{(H)} - (1, 1)$



(a)  $t^{(H)} - (1, 2)$



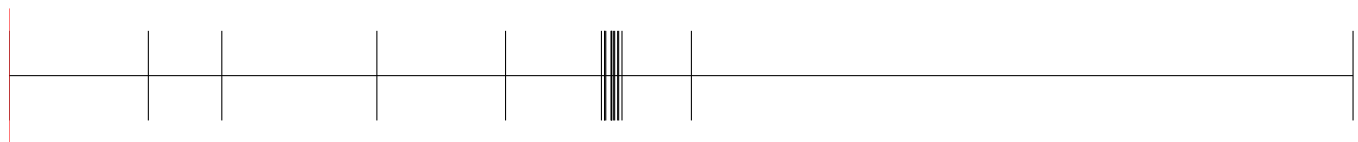
(a)  $t^{(H)} - (2, 2)$



(a)  $t^{(H)} - (1, 1, 1)$



(a)  $t^{(H)} - (1, 2, 2)$



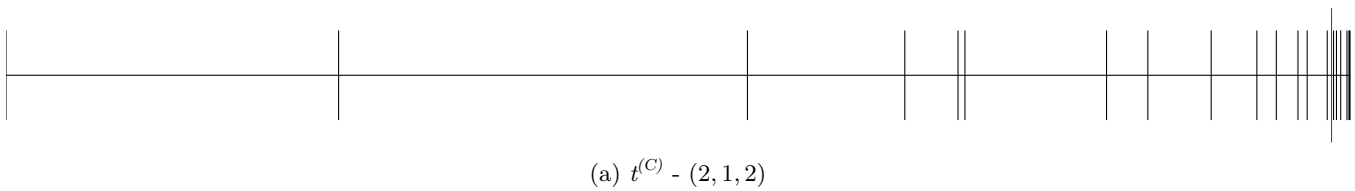
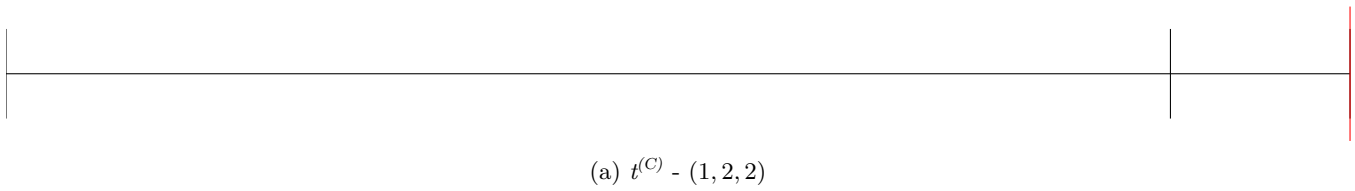
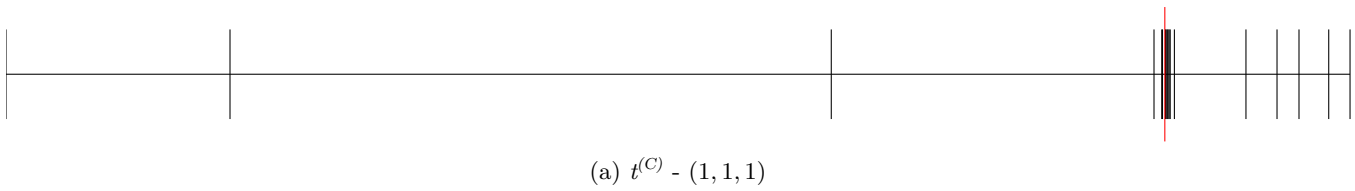
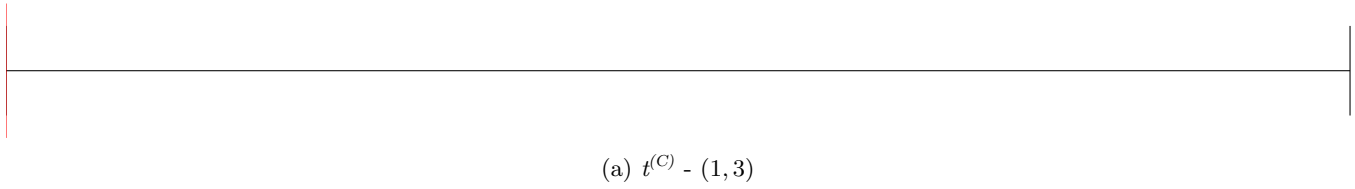
(a)  $t^{(H)} - (2, 1, 2)$



(a)  $t^{(C)} - (1, 2)$



(a)  $t^{(C)} - (2, 3)$





## 5 Area Breakpoints

### 5.1 $A$

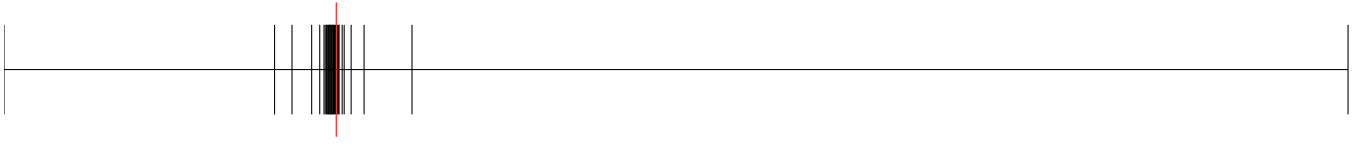
- $(1, 1, 1)$ 
  - Value: 71.627671
  - Error: 0.136630820304063
  - Relative error: 0.001903882806690
- $(1, 2, 2)$ 
  - Value: 70.566265
  - Error: 0.0000000000000966
  - Relative error: -0.000000000000014
- $(2, 1, 2)$ 
  - Value: 135.259970
  - Error: 0.195420439921946
  - Relative error: 0.001442692235919

### 5.2 $A_{CU}$

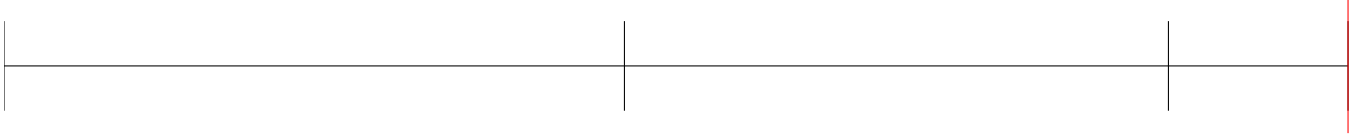
- $cu, 1$ 
  - Value: 4.600844
  - Error: 0.050719048979579
  - Relative error: 0.010903657670613
- $cu, 2$ 
  - Value: 37.950853
  - Error: 0.012025158242047
  - Relative error: 0.000316760969976

### 5.3 $A_{HU}$

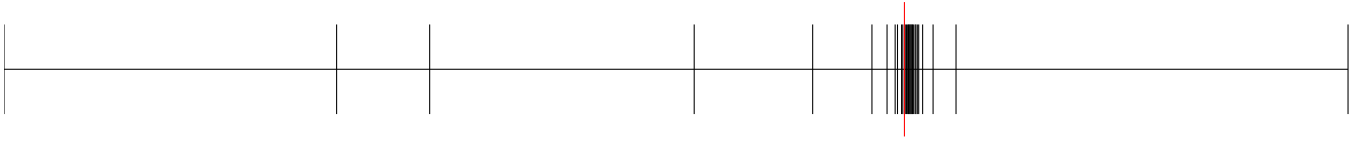
- $hu, 1$ 
  - Value: 13.363827
  - Error: 0.011446409633299
  - Relative error: 0.000855788806957



(a)  $A^\beta - (1, 1, 1)$



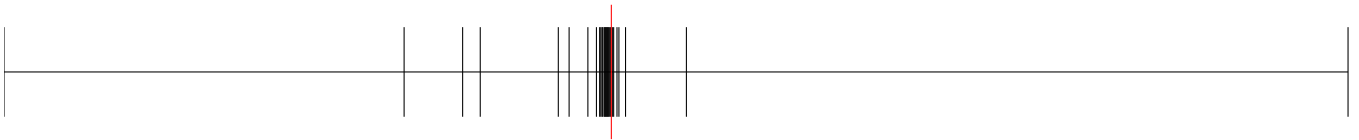
(a)  $A^\beta - (1, 2, 2)$



(a)  $A^\beta - (2, 1, 2)$



(a)  $A^\beta - cu, 1$



(a)  $A^\beta - cu, 2$



(a)  $A^\beta - hu, 1$

## 6 $A^\beta$ Breakpoints

- (1,1,1)
  - $q$ : 71.627671
  - $\hat{q}^\beta$ : 71.627671
  - $q^\beta$ : 71.627671
  - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.11420238741596, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.60541450524272, 70.65581083979713, 70.83181816751056, 70.95870595008942, 71.00935653192329, 71.18527282143235, 71.22933662920877, 71.2616536557244, **71.33750386361461**, **71.6637831223933**, 71.88123217857378, 72.01156079815767, 72.36161285231599, 560.0]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- (1,2,2)
  - $q$ : 70.566265
  - $\hat{q}^\beta$ : 70.566265
  - $q^\beta$ : 70.566265
  - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, 67.60071888621576, 67.91344539650363, 68.10969007738701, 68.48329505760876, 68.66676327863097, 68.85281320938876, 69.0314457108091, 69.22655548799992, 69.34502840170047, 69.41355730879985, 69.60484346150977, 70.08419437796795, **70.37078617087542**, **70.94081287188642**, 71.26369181826138, 72.25348351595485, 72.87506706799363, 74.6496383881616, 84.1468297226002, 390.0]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- (2,1,2)
  - $q$ : 135.259970
  - $\hat{q}^\beta$ : 135.259970
  - $q^\beta$ : 135.259970
  - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 127.07690270456857, 128.52021081477858, 131.69760457629607, 132.12228195956155, **132.5928424015674**, **135.81963163113545**, 137.02923097052596, 138.86359281280824, 139.47623233772285, 140.31685261310218, 140.7008934199032, 141.79783607898793, 142.4338700260237, 143.30274961189664, 144.67005099128556, 144.70843093515643, 147.02575214945074, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- $cu, 1$ 
  - $q$ : 4.600844
  - $\hat{q}^\beta$ : 4.600844
  - $q^\beta$ : 4.600844
  - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.0691182452509045, 4.269280092716329, 4.432280503220532, **4.440201431981878**, **4.613047202566896**, 4.707108999269609, 4.813467676447643, 4.825601156120121, 4.849719301830286, 4.888121952381175, 4.962276317638335, 4.974201802143407,

4.983938626310074, 4.993068227518925, 5.0359974771272995, 5.0418124100090695, 5.081618324442152,  
5.195781532076625, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]

– Error: 0.0000000000000000

– Relative error: 0.0000000000000000

- $cu, 2$

–  $q$ : 37.950853

–  $\hat{q}^\beta$ : 37.950853

–  $q^\beta$ : 37.950853

– Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717,  
36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, 37.52720514437563,  
37.554741031910254, 37.612770977560544, 37.65435035996605, 37.6551299822246, 37.682669717644174,  
37.71812813120871, 37.76995088018704, 37.78573239590169, 37.821651654621526, 37.88148462263016, 37.9018730666985,  
**37.9454585315849**, **38.035748486883605**, 38.10872593446926, 38.21652191748154, 38.224970554765775, 38.88483572219945,  
285.4001551947793]

– Error: 0.0000000000000000

– Relative error: 0.0000000000000000

- $hu, 1$

–  $q$ : 13.363827

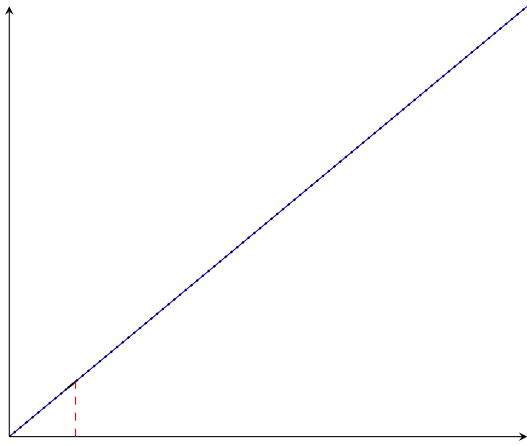
–  $\hat{q}^\beta$ : 13.363827

–  $q^\beta$ : 13.363827

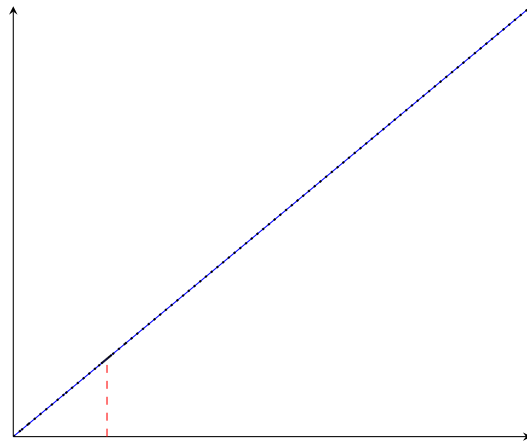
– Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861,  
13.134779411775817, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.211215176722128,  
13.213749911872785, 13.218222920629291, 13.22717920580809, 13.238585544475612, 13.244985069124171,  
13.266498143734903, 13.269097009439808, 13.294574825457861, 13.30871858348944, 13.32330648750218,  
13.32663103908035, **13.346967780023082**, **13.411355634100326**, 13.439500729482463, 13.468285806082731,  
13.498843422507207, 14.09178614751222, 237.3002543523117]

– Error: 0.0000000000000000

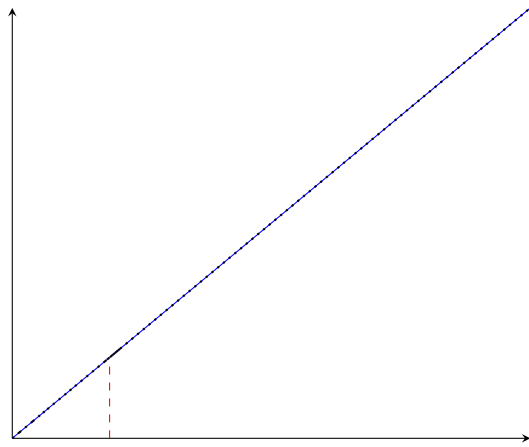
– Relative error: 0.0000000000000000



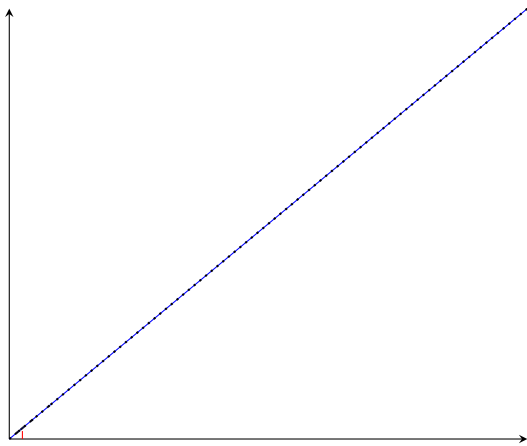
(a)  $q^\beta - (1, 1, 1)$



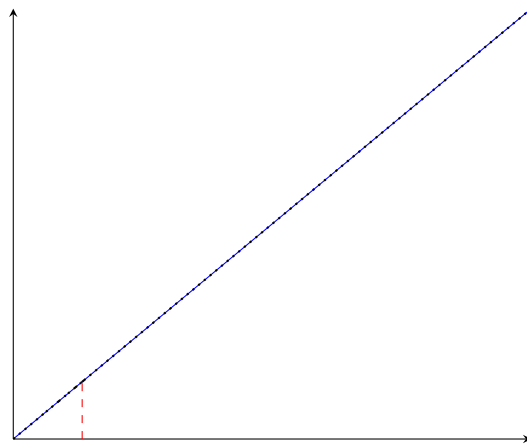
(b)  $q^\beta - (1, 2, 2)$



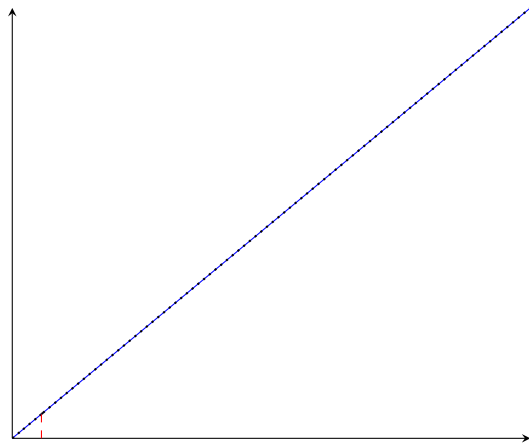
(c)  $q^\beta - (2, 1, 2)$



(a)  $q^\beta - cu, 1$



(b)  $q^\beta - cu, 2$



## 7 Other variables

| Variable          | Value             | Variable            | Value                | Variable            | Value              |
|-------------------|-------------------|---------------------|----------------------|---------------------|--------------------|
| $z_{1,1,2}$       | -0.0              | $f_{2,2,1}^C$       | 13.0                 | $b_{2,2,2}^{H,out}$ | 5332.032635825139  |
| $z_{1,2,1}$       | 0.0               | $f_{2,2,2}^C$       | 0.0                  | $b_{1,1,2}^{C,in}$  | 1978.7353704723719 |
| $z_{2,1,1}$       | -0.0              | $RecLMTD_{1,1,2}$   | 0.1                  | $b_{1,2,1}^{C,in}$  | 0.0                |
| $z_{2,2,1}$       | -0.0              | $RecLMTD_{1,2,1}$   | 0.01607047381606835  | $b_{2,1,1}^{C,in}$  | 0.0                |
| $z_{2,2,2}$       | -0.0              | $RecLMTD_{2,1,1}$   | 0.1                  | $b_{2,2,1}^{C,in}$  | 6500.0             |
| $z_{hu,2}$        | -0.0              | $RecLMTD_{2,2,1}$   | 0.006899138859007677 | $b_{2,2,2}^{C,in}$  | 0.0                |
| $A_{1,1,2}^\beta$ | 0.0               | $RecLMTD_{2,2,2}$   | 0.014501765405310972 | $b_{1,1,2}^{C,out}$ | 1978.7353704723673 |
| $A_{1,2,1}^\beta$ | 0.0               | $RecLMTD_{hu,2}$    | 0.005549329826813255 | $b_{1,2,1}^{C,out}$ | 0.0                |
| $A_{2,1,1}^\beta$ | 0.0               | $q_{1,1,2}$         | 0.0                  | $b_{2,1,1}^{C,out}$ | 0.0                |
| $A_{2,2,1}^\beta$ | 0.0               | $q_{1,2,1}$         | 0.0                  | $b_{2,2,1}^{C,out}$ | 6500.0             |
| $A_{2,2,2}^\beta$ | 0.0               | $q_{2,1,1}$         | 0.0                  | $b_{2,2,2}^{C,out}$ | 0.0                |
| $A_{hu,2}^\beta$  | 0.0               | $q_{2,2,1}$         | 0.0                  |                     |                    |
| $A_{1,1,2}$       | 0.0               | $q_{2,2,2}$         | 0.0                  |                     |                    |
| $A_{1,2,1}$       | 0.0               | $q_{hu,2}$          | 0.0                  |                     |                    |
| $A_{2,1,1}$       | 0.0               | $t_{1,1,2}^H$       | 370.0                |                     |                    |
| $A_{2,2,1}$       | 0.0               | $t_{1,2,1}^H$       | 650.0                |                     |                    |
| $A_{2,2,2}$       | 0.0               | $t_{2,1,1}^H$       | 590.0                |                     |                    |
| $A_{hu,2}$        | 0.0               | $t_{2,2,1}^H$       | 590.0                |                     |                    |
| $dt_{1,1,3}$      | 10.0              | $t_{2,2,2}^H$       | 590.0                |                     |                    |
| $dt_{1,2,1}$      | 150.0             | $t_{1,1,2}^C$       | 410.0                |                     |                    |
| $dt_{2,1,1}$      | 10.0              | $t_{1,2,1}^C$       | 350.0                |                     |                    |
| $dt_{2,2,1}$      | 220.0             | $t_{2,1,1}^C$       | 410.0                |                     |                    |
| $dt_{2,2,2}$      | 64.40191065600966 | $t_{2,2,1}^C$       | 500.0                |                     |                    |
| $dt_{2,2,3}$      | 70.0              | $t_{2,2,2}^C$       | 440.01120358475225   |                     |                    |
| $dt_{hu,2}$       | 180.0             | $b_{1,1,2}^{H,in}$  | 0.0                  |                     |                    |
| $f_{1,1,2}^H$     | 0.0               | $b_{1,2,1}^{H,in}$  | 3412.00374034169     |                     |                    |
| $f_{1,2,1}^H$     | 5.2492365236026   | $b_{2,1,1}^{H,in}$  | 11800.0              |                     |                    |
| $f_{2,1,1}^H$     | 20.0              | $b_{2,2,1}^{H,in}$  | 0.0                  |                     |                    |
| $f_{2,2,1}^H$     | 0.0               | $b_{2,2,2}^{H,in}$  | 5332.032635825154    |                     |                    |
| $f_{2,2,2}^H$     | 9.037343450551141 | $b_{1,1,2}^{H,out}$ | 0.0                  |                     |                    |
| $f_{1,1,2}^C$     | 4.826183830420419 | $b_{1,2,1}^{H,out}$ | 3412.003740341714    |                     |                    |
| $f_{1,2,1}^C$     | 0.0               | $b_{2,1,1}^{H,out}$ | 11800.0              |                     |                    |
| $f_{2,1,1}^C$     | 0.0               | $b_{2,2,1}^{H,out}$ | 0.0                  |                     |                    |